Sapienza Università di Roma

Department of Mechanical and Aerospace Engineering

Homework 3 — Attitude Control of a Launch Vehicle in Atmospheric Flight

Dynamics and Control of Launch Vehicles

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1 Introduction

This report addresses the design of an attitude control system for a launch vehicle (LV) during the atmospheric phase of flight, at the instant of maximum dynamic pressure ($t = 72$ s, max-$\bar{q}$). This is the hardest point to control: the aerodynamic moment destabilises the airframe and the wind disturbance is largest. I design the controller with classical frequency-domain tools, using the Nichols chart as the main stability indicator, following the short-period pitch-plane model of Greensite [1].

1.1 Design methodology: frozen coefficients at max-$\bar{q}$

Launch-vehicle attitude control in atmospheric flight is classically handled by the **time-slice** (frozen-coefficient) method: the slowly varying mass, inertia and aerodynamic coefficients are frozen at a representative instant, a linear time-invariant short-period model is built and analysed at that operating point, and the controller is then validated against time-varying simulations [1, Sec. 2]. The main sizing disturbance is the wind profile, which drives both the trajectory dispersion and the structural bending loads—hence the gust analysis of Tasks 1 and 2. I pick $t = 72$ s because it is the load- and stability-critical slice: the aerodynamic destabilising moment and the wind-induced angle of attack are both near their maxima. The catch is that point-wise Nichols margins certify only the frozen model and do not by themselves guarantee stability of the underlying time-varying system; this is what motivates the parametric robustness study of Task 3 and, eventually, gain scheduling along the trajectory.

1.2 Plant model

The pitch-plane short-period dynamics are described by the linear time-invariant model (assignment Eq. 1), with state vector $\mathbf{x} = [\,z,\ \dot{z},\ \theta,\ \dot{\theta},\ \eta,\ \dot{\eta}\,]^\top$ (lateral drift, pitch attitude and first bending mode), control input δ (TVC deflection) and disturbance α_w (wind angle of attack):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & a_1 & a_1 V + a_4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & A_6/V & A_6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -\omega_1^2 & -2\zeta_1\omega_1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ a_3 \\ 0 \\ K_1 \\ 0 \\ -\phi_{1,TVC} T_c \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -a_1 V \\ 0 \\ -A_6 \\ 0 \\ 0 \end{bmatrix} \alpha_w. \quad (1)$$

The pitch channel is the core of the problem and follows from a moment balance about the centre of mass. The aerodynamic moment $N_\alpha l_\alpha$ acts on the local incidence—built from the pitch attitude θ, the plunging term $\dot{z}/V$ and the wind angle α_w—while the gimballed thrust contributes a control moment $T_c l_c\, \delta$; dividing the balance $I_{yy}\ddot{\theta} = N_\alpha l_\alpha(\theta + \dot{z}/V - \alpha_w) + T_c l_c\, \delta$ by I_{yy} reproduces row 4 of Eq. (1),

$$\ddot{\theta} = A_6\, \theta + \frac{A_6}{V}\, \dot{z} + K_1\, \delta - A_6\, \alpha_w, \tag{2}$$

and identifies the two structural coefficients: the **aerodynamic moment** $A_6 = N_\alpha l_\alpha / I_{yy} > 0$ (denoted μ_α) and the **control effectiveness** $K_1 = T_c l_c / I_{yy}$ (μ_c). With $A_6 > 0$ the airframe transfer function $\theta/\delta = K_1/(s^2 - A_6)$ carries an open-loop pole at $s = +\sqrt{A_6} = +1.84$ rad/s in the right half-plane: the **statically unstable** airframe of a vehicle whose centre of pressure lies ahead of its centre of mass, for which feedback is mandatory [2, Lecture 15]. The TVC coefficient K_1 sets the control authority available to counter it.

The sensed quantities are produced by the inertial unit (assignment Eq. 2), which couples the bending coordinate η into the gyro and accelerometer channels through $\sigma_{1,ins}$ and $\phi_{1,ins}$:

$$\theta_m = \theta + \sigma_{1,ins}\, \eta, \quad \dot{\theta}_m = \dot{\theta} + \sigma_{1,ins}\, \dot{\eta}, \quad z_m = z - \phi_{1,ins}\, \eta, \quad \dot{z}_m = \dot{z} - \phi_{1,ins}\, \dot{\eta}. \tag{3}$$

This contamination is exactly what makes the lightly damped first bending mode—written with $\omega_1 \equiv \omega_{BM} = 18.9$ rad/s and $\zeta_1 \equiv \zeta_{BM} = 0.005$ in Eq. (1)—visible to the controller and drives the need for a dedicated filter (Task 2).

The actuator is the second-order TVC of Eq. 3,

$$W_{TVC}(s) = \frac{\omega_{TVC}^2}{s^2 + 2\zeta_{TVC}\omega_{TVC} s + \omega_{TVC}^2}, \qquad \omega_{TVC} = 70 \text{ rad/s},\ \zeta_{TVC} = 0.7, \tag{4}$$

in series with a pure transport delay $\tau = 20$ ms. The delay is approximated by a third-order Padé rational function $e^{-\tau s} \approx P_3(s)$, the diagonal [3/3] member of the family obtained by matching the Taylor expansion of $D(s)\, e^{-\tau s} = N(s)$ to the highest possible order; the illustrative second-order form is

$$e^{-\tau s} \approx \frac{12 - 6\, \tau s + (\tau s)^2}{12 + 6\, \tau s + (\tau s)^2}, \tag{5}$$

an **all-pass** rational function ($|P(j\omega)| = 1$ for all ω) that reproduces the phase lag of the delay while leaving the loop gain unchanged. I keep order three, matching the implementation, so that the approximation stays accurate up to the bending frequency, where $\omega_{BM}\tau \approx 0.38$ is no longer negligible. Parameters at $t = 72$ s are read from the reference data set `GreensiteLPV_DATA.mat` (cross-checked against Table 1 of the assignment); the salient values are reported in Table 1.

Table 1: Pitch-plane LV parameters at the max-$\bar{q}$ condition ($t = 72$ s).

Quantity	Symbol	Value
Aerodynamic moment	$A_6 = \mu_\alpha$	3.382 s^{-2}
Control effectiveness	$K_1 = \mu_c$	4.565 s^{-2}
Drift coefficients	a_1, a_3, a_4	-0.0154, 20.61, -27.27 s^{-2}
Relative velocity	V	937.7 m/s
Bending mode	ω_{BM}, ζ_{BM}	18.9 rad/s, 0.005
INS coupling	$\phi_{1,ins}$ [–], $\sigma_{1,ins}$ [rad/m]	0.8, 0.178
TVC actuator	$\omega_{TVC}, \zeta_{TVC}, \tau$	70 rad/s, 0.7, 20 ms

1.3 Control architecture

A proportional–derivative attitude law is adopted, augmented by a weak feedback on the lateral-drift channel:

$$\delta_{cmd} = K_{P,\theta}\,(\theta_{ref} - \theta_m) - K_{D,\theta}\,\dot{\theta}_m - K_{P,z}\,z_m - K_{D,z}\,\dot{z}_m, \tag{6}$$

with $K_{P,z}, K_{D,z}$ small and negative, of order 10^{-3}, as recommended in the assignment guidelines. Closing the loop also on $(z_m, \dot{z}_m)$ trades between two classical objectives in wind: containing the lateral drift (drift minimum) and containing the angle of attack $\alpha = \theta + \dot{z}/V - \alpha_w$, hence the structural load $\bar{q}\,\alpha$ (load minimum) [2, Lecture 17] — the binding concern at the max-$\bar{q}$ condition studied here. The sign of α_w is the one implied by Eq. (1): α is the incidence measured against the *air-relative* velocity, so a lateral wind subtracts from the vehicle's own $\dot{z}$. Note that no weathercock stability is available to relieve the load — with $A_6 > 0$ the centre of pressure sits *ahead* of the centre of mass and the aerodynamic moment is divergent, so an attitude-hold law is if anything load aggravating (Task 1). The weak negative gains close the free integrator in lateral position and keep the drift bounded; I therefore monitor both z and $\bar{q}\,\alpha$ in the gust simulations. The actual deflection follows from the actuator and the bending filter $H_x(s)$ (Task 2): $\delta = W_{TVC}(s)\,e^{-\tau s}\,H_x(s)\,\delta_{cmd}$. The complete loop is sketched in Figure 1.

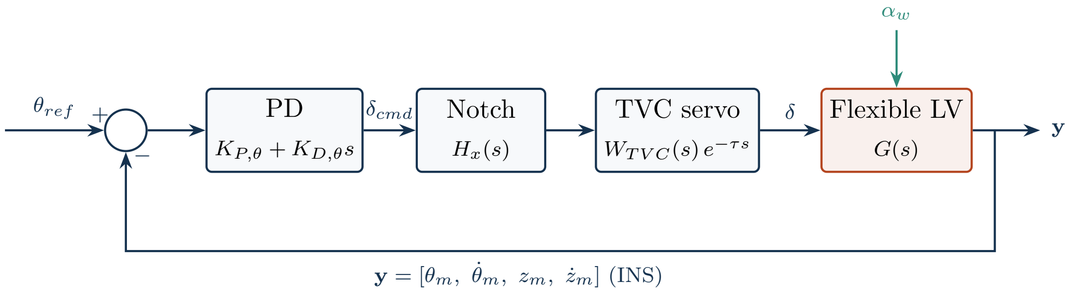


Figure 1: Full attitude-control loop (Tasks 2–3): the PD law, the bending notch H_x, the TVC servo with transport delay, and the flexible airframe forced by the wind angle of attack α_w. The INS measurement vector $\mathbf{y}$ closes the loop; its lateral-drift channels carry the weak $K_{P,z}, K_{D,z}$ feedback of Eq. (6). Task 1 uses the same loop with $H_x = 1$, $W_{TVC} = 1$ and the rigid plant (Figure 4).

1.4 A note on conditional stability

Because the open-loop airframe carries a single right-half-plane pole (the +1.84 rad/s aerodynamic pole $s = +\sqrt{\mu_\alpha}$), the loop is **conditionally stable**: the proportional gain must be large enough to overcome the static aerodynamic instability, $K_{P,\theta} > \mu_\alpha/\mu_c$, so stability is lost if the loop gain is reduced below this threshold, not only if it is increased [2, Lecture 15]. An aerodynamically unstable vehicle driven through a second-order (or higher) actuator therefore has two gain margins—a low-frequency aerodynamic margin (reduce gain → unstable) and a high-frequency rigid-body margin (increase gain → unstable)—which on the Nichols chart show up as two 0 dB crossings of the −180° line [2, Lecture 16]. The margins are therefore read on the *full* open loop and classified by frequency band, rather than taken from `margin`'s default crossover: the aerodynamic gain margin at the low-frequency −180° crossing (a gain-reduction margin, $|GM_a| \approx 6$ dB), the rigid-body phase margin at the gain crossover ($PM \approx 30°$), and—once the actuator and bending are present (Task 2)—the high-frequency rigid-body gain margin and the bending Flex margins. This is standard practice for the attitude control of aerodynamically unstable launchers [3].

The same conditional stability is visible in the s-plane: the root locus of the rigid loop (Figure 2) starts at the open-loop poles, one of which — the aerodynamic pole $+\sqrt{\mu_\alpha}$ — lies in the right half-plane, and, as the loop gain grows, that pole is drawn across the imaginary axis into the left half-plane. Stabilising the airframe therefore demands a *minimum* loop gain, $K_{P,\theta} > \mu_\alpha/\mu_c$; the upper bound that closes the finite stable band comes from the actuator lag and is read on the Nichols chart.

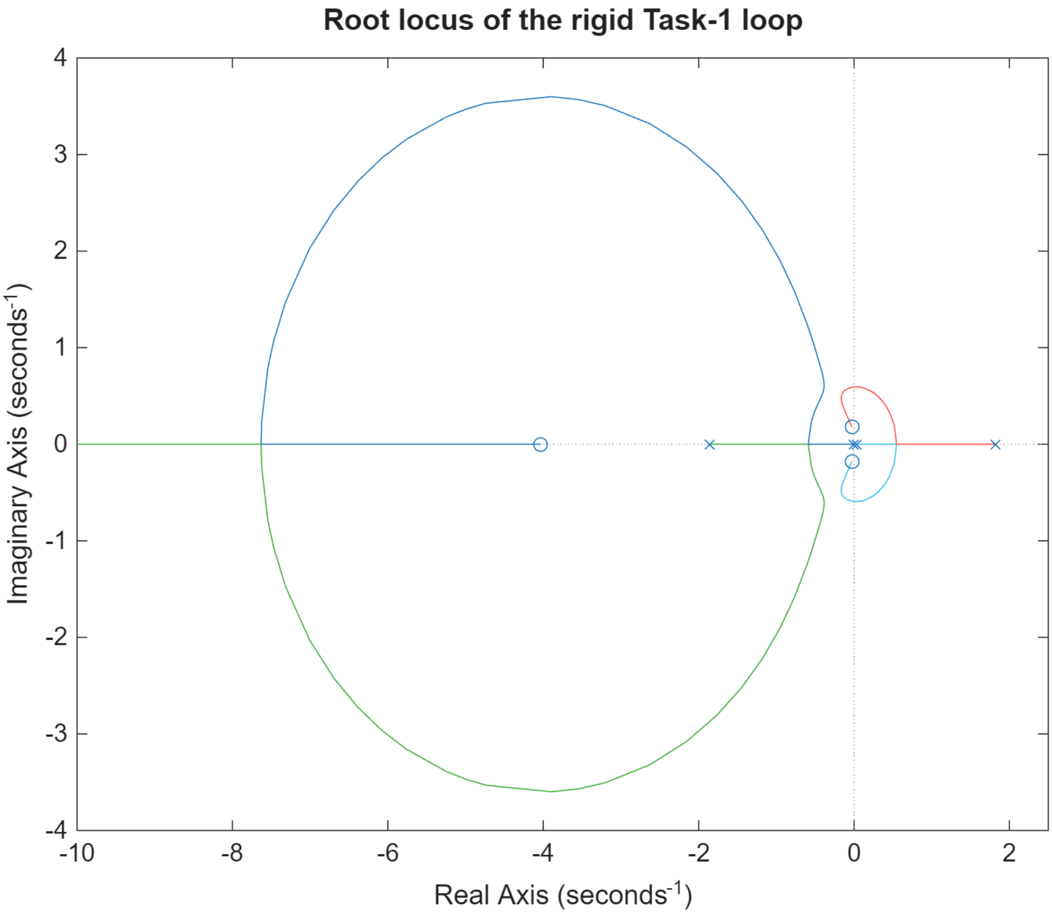


Figure 2: Conditional stability on the root locus of the rigid Task-1 loop (built-in `rlocus`). Following the usual convention, $\times$ marks the open-loop poles (where the branches start, $k = 0$) and $\circ$ the open-loop zeros (where they end, $k \to \infty$); the branches trace the closed-loop poles as the loop gain grows. As the loop gain grows the unstable aerodynamic pole at $+\sqrt{\mu_\alpha} = +1.84$ rad/s ($\times$ in the right half-plane) is drawn into the left half-plane, so the closed loop is stable at the design gain; the drift mode leaves the real axis into the lightly damped pair near the imaginary axis. Stability therefore requires a *minimum* loop gain, $K_{P,\theta} > \mu_\alpha/\mu_c = 0.74$ (the low-frequency aerodynamic margin); the high-frequency margin set by the actuator lag is the second 0 dB crossing on the Nichols chart.

2 Task 1 — Rigid Launch Vehicle

Figure 3 sketches the design pipeline of `main_task1.m`, which Tasks 2 and 3 reuse with richer plant and actuator models.

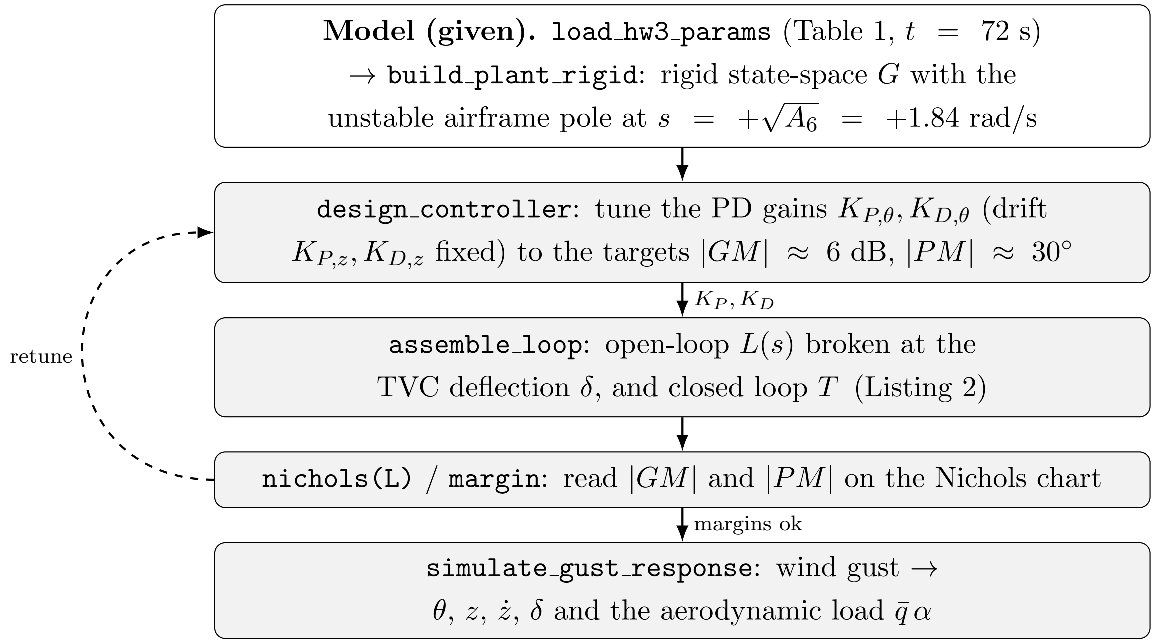


Figure 3: Task 1 design pipeline (`main_task1.m`). Tasks 2 and 3 reuse it with the full flexible plant and the TVC+delay+notch chain `Wact`. Listing numbers refer to Appendix A.

2.1 Rigid model and loop

Neglecting the bending mode, the plant reduces to the four states $[z, \dot{z}, \theta, \dot{\theta}]$ (`build_plant_rigid.m`, Listing 1) and the actuator is taken as ideal: $W_{TVC}(s) = 1$ and no transport delay. The open-loop transfer broken at the TVC deflection, in the $1 + L$ convention, is obtained by closing the static PD law of Eq. (6) around the plant (`assemble_loop`, Listing 2):

$$L(s) = K_{P,\theta}\, G_{\delta\to\theta}(s) + K_{D,\theta}\, G_{\delta\to\dot{\theta}}(s) + K_{P,z}\, G_{\delta\to z}(s) + K_{D,z}\, G_{\delta\to\dot{z}}(s), \tag{7}$$

where $G_{\delta\to\bullet}$ are the plant channels from δ to each measurement. The dominant term is the pitch loop $L_\theta(s) = (K_{P,\theta} + K_{D,\theta}s)\, K_1/(s^2 - A_6)$, whose closed-loop characteristic polynomial $s^2 + K_1K_{D,\theta}\, s + (K_1K_{P,\theta} - A_6)$ is stable only for $K_{P,\theta} > A_6/K_1 = 0.74$ and $K_{D,\theta} > 0$, so the proportional gain has to overcome the aerodynamic instability. The rigid loop is sketched in Figure 4.

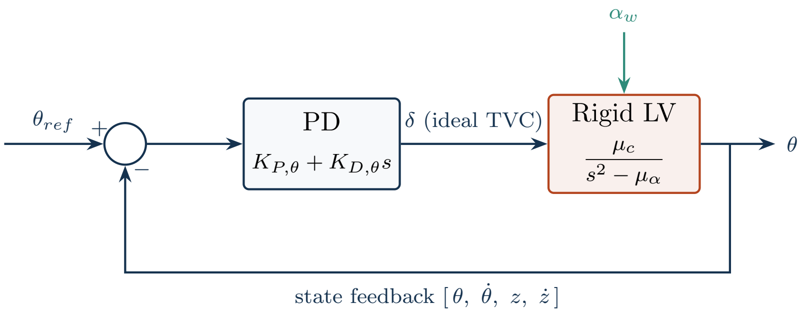


Figure 4: Task 1 rigid loop: the PD law drives the airframe through an ideal actuator ($W_{TVC} = 1$, no delay or notch). The pitch channel shown here, $\mu_c/(s^2 - \mu_\alpha)$, carries the unstable pole at $+\sqrt{\mu_\alpha}$; the full loop with the actuator chain is Figure 1.

The gain and phase margins quoted below are read on the *loop transfer* $L(s)$, obtained by **opening this loop at the TVC deflection** δ (Figure 5): δ is the single actuator signal, so cutting the wire there, injecting a test signal δ_{in} into the plant and reading the returning δ_{out} at the actuator output gives a scalar $L(s) = \delta_{out}/\delta_{in}$. This is why the Nichols charts are labelled "from δ to δ": input and output sit at the same break point. Closed-loop poles are the roots of $1 + L = 0$.

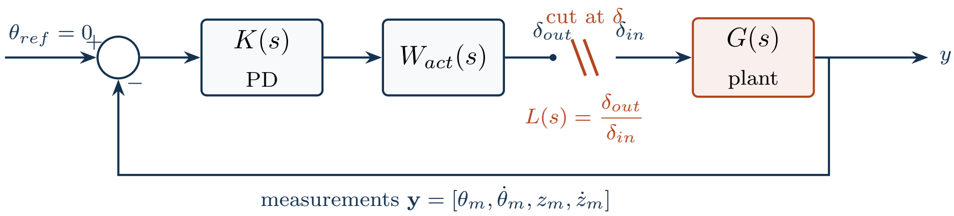


Figure 5: Where the loop is opened. The loop transfer $L(s)$ used for the Nichols charts and the gain/phase margins is formed by cutting the loop at the actuator output δ (the plant input): a test δ_{in} is injected into the plant and the returning δ_{out} is read back around the loop, so $L(s) = \delta_{out}/\delta_{in}$ is a single scalar transfer "from δ to δ". Here $W_{act} = 1$ (ideal actuator, Task 1); for Tasks 2–3 $W_{act} = H_x(s)\, W_{TVC}(s)\, e^{-\tau s}$ (notch, TVC servo and delay).

2.2 Tuning

The pitch gains follow the classical launcher design of the course notes [2, Lecture 16]. Starting from the canonical closed form on the decoupled rotational dynamics $\ddot{\theta} = A_6\theta + K_1\delta$,

$$K_{P,\theta}^{(0)} = \frac{2A_6}{K_1}, \qquad K_{D,\theta}^{(0)} = \frac{\sqrt{A_6}}{K_1}, \tag{8}$$

which place the pitch pair at $\omega_c = \sqrt{A_6}$, $\zeta = 0.5$ and give a 6 dB gain margin and 30° phase margin *on that decoupled loop*. On the full four-state loop the lateral-drift feedback erodes the aerodynamic gain margin (the 6 dB drops to about 4 dB), so a two-parameter `fminsearch` retunes $(K_{P,\theta}, K_{D,\theta})$ on the *full* loop until the frequency-classified aerodynamic gain margin and rigid-body phase margin meet the assignment targets. The lateral gains stay fixed at $K_{P,z} = K_{D,z} = -10^{-3}$ (load relief, not a stability margin). The result is

$$K_{P,\theta} = 1.78, \quad K_{D,\theta} = 0.44, \quad K_{P,z} = K_{D,z} = -10^{-3}, \tag{9}$$

with, read on the full loop, an aerodynamic gain margin $|GM_a| = 6.0$ dB at 0.59 rad/s, a rigid-body phase margin $PM = 30°$ at 2.45 rad/s, and a delay margin of 213 ms. In the pole-placement language of the course notes [2, Lecture 15] these gains place the pitch pair at $\omega_c = \sqrt{K_1 K_{P,\theta} - A_6} = 2.18$ rad/s — inside the usual 1–4 rad/s control band — with damping $\zeta = K_1 K_{D,\theta}/(2\omega_c) = 0.46$, in the standard 0.4–0.7 range. These relations invert the closed-loop characteristic polynomial, matching $s^2 + K_1 K_{D,\theta}\, s + (K_1 K_{P,\theta} - A_6)$ to $s^2 + 2\zeta\omega_c s + \omega_c^2$:

$$K_{P,\theta} = \frac{\omega_c^2 + A_6}{K_1}, \qquad K_{D,\theta} = \frac{2\zeta\omega_c}{K_1}. \tag{10}$$

Reading the margins. The margins are read on the *full* open loop and classified by frequency band, not taken from `margin`'s default crossover [3]. Because the lateral-drift position feedback carries a free integrator ($z/\delta \to \infty$ at DC), the Nichols curve comes from the top (Figure 6): the aerodynamic gain margin is the low-frequency gain-reduction margin, read where the curve crosses the critical point — $(-180°, 0\text{ dB})$, from $1 + L = 0$ — at $|GM_a| = 6$ dB, and the rigid-body phase margin is read at the 2.45 rad/s gain crossover. With the ideal Task-1 actuator these are the only rigid-body margins; the high-frequency rigid-body gain margin (a gain-increase margin set by the actuator lag) and the bending Flex margins appear once the TVC servo, transport delay and notch are added in Task 2. Crossings below ~ 0.5 rad/s are lateral-drift artifacts, not rigid-body margins.

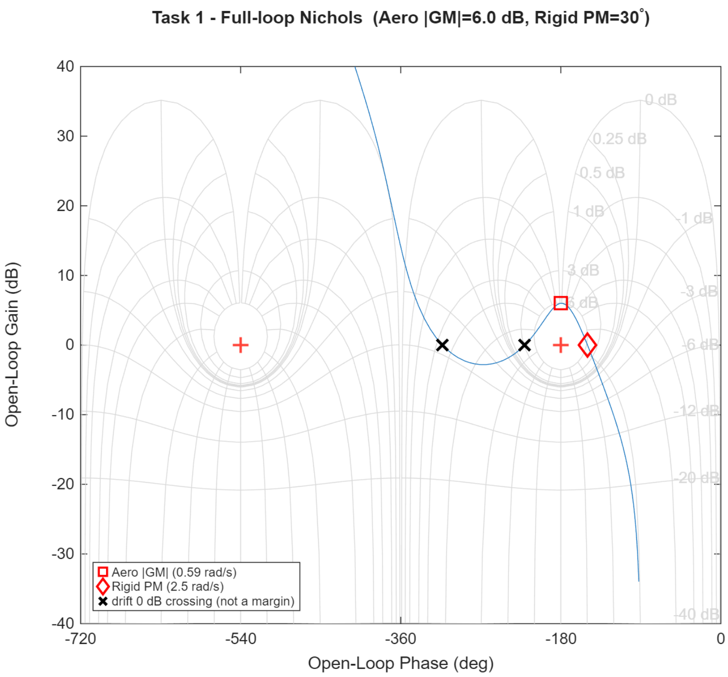


Figure 6: Task 1 — full-loop Nichols; the critical point is $(-180°, 0\text{ dB})$, from $1 + L = 0$. The loop comes from the top (lateral-drift integrator); the design meets $|GM_a| \approx 6$ dB (aerodynamic, at $-180°$) and $PM \approx 30°$ (rigid-body, at the 2.45 rad/s crossover).

In the s-plane the same design reads as a pole placement (Figure 7): the PD gains pull the unstable $+1.84$ rad/s aerodynamic pole into the left half-plane, leaving a well-damped fast pitch pair ($-0.95 \pm 1.90i$, $\zeta \approx 0.45$) and a slow, lightly damped drift pair ($-0.06 \pm 0.23i$) close to the imaginary axis — the stability margin the Nichols chart quantifies.

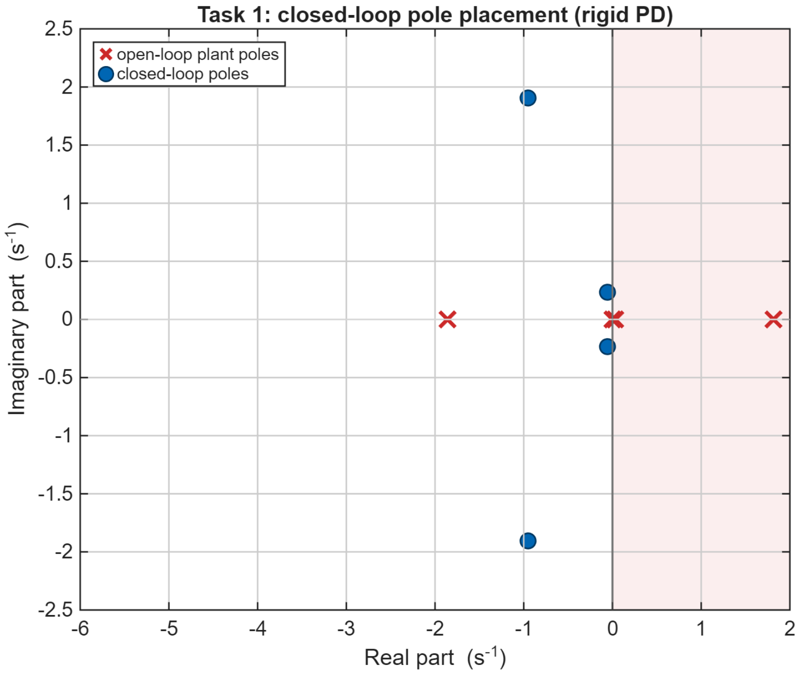


Figure 7: Task 1 — open-loop plant poles (crosses, one in the right half-plane) against the closed-loop poles (circles): the rigid PD stabilises the airframe.

2.3 Wind-gust response

The closed loop is excited by a deterministic "1-cosine" wind gust whose amplitude ($V_g = 6.4$ m/s) is taken from the reference dry-wind dispersion (`drywind.mat`, severe regime) at the current altitude, converted to a wind angle of attack $\alpha_w = v_w/V$ with a peak of 0.39°. The four key time histories requested by the assignment are shown in Figure 8: the pitch attitude peaks at 0.26°, the TVC deflection at 0.53°, and the lateral drift peaks at 2.3 m before the slow, lightly damped drift mode ($\zeta \approx 0.23$, $\omega_n \approx 0.24$ rad/s) returns it to trim. All four states decay back to zero within about 80 s — confirming the loop is asymptotically stable — and the actuator effort stays small.

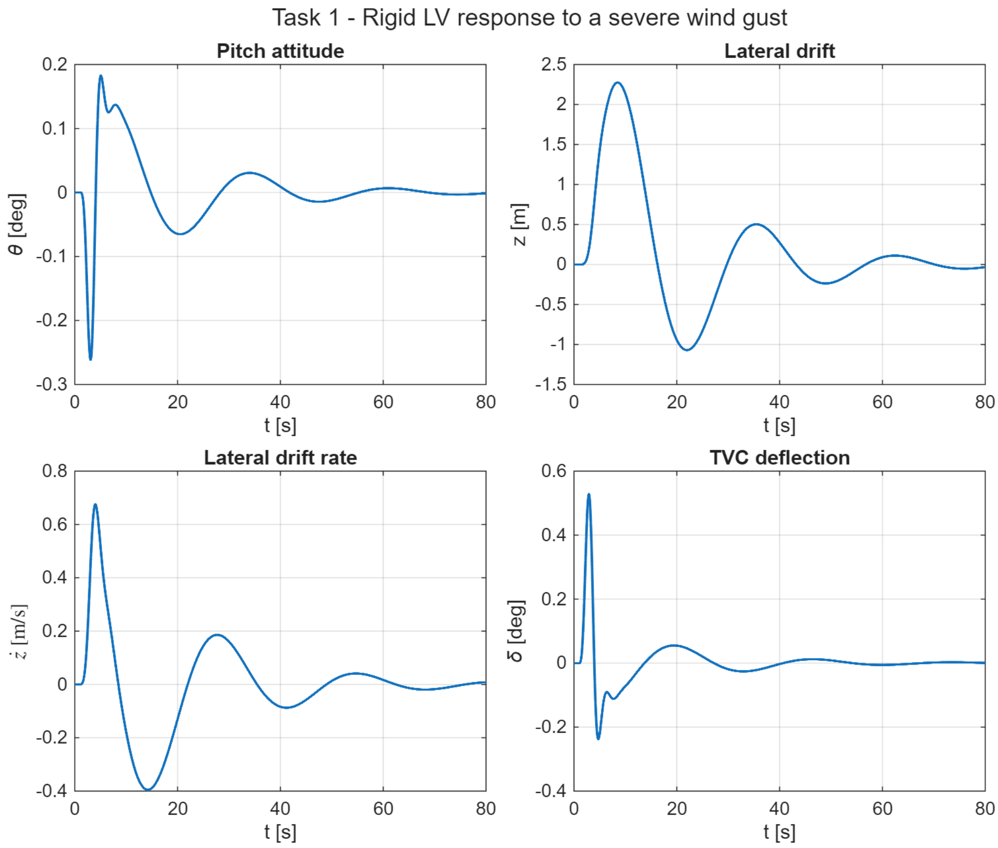


Figure 8: Task 1 — response to a severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

Since max-$\bar{q}$ is the load-critical point of the trajectory, the total angle of attack $\alpha = \theta + \dot{z}/V - \alpha_w$ and the structural-load indicator $\bar{q}\,\alpha$ are also monitored (Figure 9). The minus sign is the model's own: α is the incidence seen by the *air-relative* velocity, and it is the quantity that multiplies A_6 and $a_1 V$ in Eq. (1), whose wind column is $\mathbf{B}_w = [0,\ -a_1 V,\ 0,\ -A_6,\ 0,\ 0]^\top$. To hold attitude against the gust the loop pitches the vehicle into the relative wind ($\theta \to -0.26°$ at the gust peak), and that contribution *adds* to the wind's own rather than cancelling it: the peak total incidence, 0.577°, therefore *exceeds* the 0.39° of the gust alone. With $\bar{q} = 81$ kPa the peak load indicator is $\bar{q}\,\alpha = 47$ kPa deg. A pure attitude-hold law is thus mildly *load aggravating* — which is precisely why flight designs add an explicit load-relief term. The incidence nevertheless stays an order of magnitude below the few-degree limit typical of a slender launcher, so at this gust level the design is not load critical.

This drift-versus-load tension is the standard trade in launch-vehicle wind control [1, Sec. 3.1]. Applying the final-value theorem to a *sustained* step gust, with the attitude loop holding $\theta_{ss} = 0$, the aerodynamic moment can only vanish if the incidence does: $\alpha_{ss} = 0$, which forces a steady drift rate $\dot{z}_{ss} = +V\,\alpha_w$. The vehicle is simply blown downwind until it flies *with* the air — it nulls the relative incidence but not the dispersion. The 1-cosine gust of Figure 8 is instead a transient pulse, so once it has passed the stable loop returns the drift to zero, but only through the slow, lightly damped mode, which is why the settling takes $\sim$80 s.

The weak negative drift gains are, first of all, a *stability* requirement rather than a load-relief device. Lateral position has no restoring force: the rigid airframe carries a free integrator in z, and with $K_{P,z} = K_{D,z} = 0$ the closed loop keeps a pole exactly at the origin — poles $\{0, -0.076, -0.98\pm1.93\mathrm{i}\}$, marginally stable, with z never returning to zero. The -10^{-3} gains move that pair to $-0.056 \pm 0.233\mathrm{i}$, cutting the peak drift from 9.5 m to 2.3 m while changing the peak incidence by barely 1% (0.584° → 0.577°). They buy drift control, not load relief. The **load-minimum** condition of the classical literature ($K_{P,\theta} = 0$, so that $\alpha_{ss} = 0$) is moreover inadmissible on this airframe: $K_{P,\theta}$ is precisely what stabilises the $+\sqrt{A_6}$ aerodynamic root, and setting it to zero makes the loop diverge. Genuine load relief on an unstable launcher must therefore come from an added feedback, not from detuning the attitude channel: the classical options are a direct incidence term $K_\alpha \alpha$ (needs an air-data measurement or estimate), a lateral-acceleration term $K_a \ddot{z}$, or larger translational gains on $(z, \dot{z})$ — the latter being the structure used here, but sized for drift control rather than load relief.

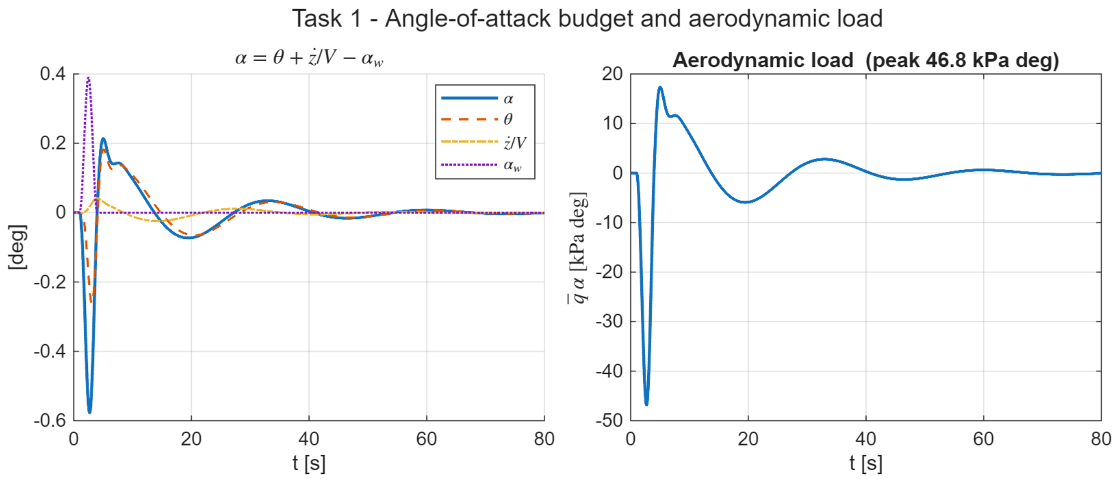


Figure 9: Task 1 — angle-of-attack budget $\alpha = \theta + \dot{z}/V - \alpha_w$ (left) and aerodynamic load indicator $\bar{q}\,\alpha$ (right) during the severe gust. The attitude loop pitches into the relative wind, so θ and $-\alpha_w$ add: the peak incidence (0.577°) exceeds the gust alone (0.39°).

3 Task 2 — Full Launch Vehicle Model

Task 2 reintroduces the full dynamics of Eq. (1): the lightly damped first bending mode, the second-order TVC actuator of Eq. (4) with its 20 ms delay, and the INS measurement model of Eq. (3). I build the design up incrementally, choosing the bending filter first and then — unlike a naive reuse of the Task-1 gains — **re-tuning the PD on the full loop**: the Task-1 PD was placed on the *ideal*-actuator loop, and once the real TVC servo, transport delay and notch are in the path their phase lag collapses the rigid phase margin, so the gains must be re-tuned to recover it (D'Antuono: the TVC actuator dynamics must be taken into account when placing the PD [3]).

3.1 Step B — uncompensated full model

Adding the TVC, delay and bending mode to the Task 1 controller leaves the low-frequency margins almost untouched, but the bending resonance shows up in the loop at $|L(\omega_{BM})| = +29$ dB. With a damping of only $\zeta_{BM} = 0.005$ this peak crosses the critical point and the closed loop goes **unstable** (a closed-loop pole at $\mathrm{Re}(s) = +1.6$).

3.2 Step C — bending filter trade study

The guideline filter of Eq. 4 admits two readings: as printed (negative numerator damping, a non-minimum-phase **lead-lag** that phase-stabilises the resonance by rotating it on the Nichols chart) or in its minimum-phase variant (a band-reject **notch** that gain-stabilises the resonance by driving it below 0 dB) [2, Lecture 18]. Instead of picking one up front, I compare four candidates at the fixed Task-1 gains (all built by `build_notch_filter.m`, Listing 3); Table 2 collects the resulting margins, classified by frequency band (aerodynamic gain margin, rigid-body phase and gain margin, delay margin on top of the 20 ms already modelled, and the residual loop gain at the bending frequency). The trade fixes the filter *architecture*; the PD is re-tuned afterwards.

- **C-LL — Eq. 4 lead-lag alone.** Swept over the ranges suggested in the assignment ($\zeta_{x,N} \in [0.1, 0.3]$, $\zeta_{x,D} \in [0.4, 0.6]$, $\omega_x = \omega_{BM} \pm 4$ rad/s, 75 combinations), 45/75 now stabilise the loop — the milder +29 dB resonance (a consequence of the smaller Task-1 derivative gain) is within reach of pure phase rotation. But the best combination ($\omega_x = 18.9$, $\zeta_{x,N} = 0.30$, $\zeta_{x,D} = 0.60$) is only barely stable ($\mathrm{Re}(s) = -0.06$) with a thin rigid phase margin (11°): phase stabilisation of the first mode leaves almost no room.
- **C-N — deep minimum-phase notch.** A deep min-phase notch ($\zeta_{x,N} = 0.002$,

$\zeta_{x,D} = 0.7$) gain-stabilises the resonance, driving $|L(\omega_{BM})|$ to -22 dB. The depth follows the closed-form attenuation of a second-order notch, $K_{\text{notch}} = 20\log_{10}(\zeta_{x,N}/\zeta_{x,D}) = 20\log_{10}(0.002/0.7) \approx -51$ dB [2, Lecture 18], which is why the sizing has to leave the assignment's suggested $\zeta_{x,N} \in [0.1, 0.3]$ range. I keep the null narrow because the bending frequency sits only a decade above the rigid crossover: a wide band-reject section would leak phase lag into the conditionally-stable low-frequency region. This is the retained architecture.

- **C-T — notch triplet.** The notch-triplet recipe against bending-frequency uncertainty places three cascaded notches at $\{0.9,\ 1.0,\ 1.1\}\,\omega_{BM}$ [2, Lecture 18]. Sized to the same depth, the three wide ($\zeta_{x,D} = 0.7$) sections pile up about 30° of phase lag at the rigid crossover and the loop goes **unstable** at nominal: the first mode is too close to the control band to blanket-notch (triplets belong to the higher, well-separated modes).
- **C-NLL — notch + lead-lag.** Reading the guideline literally ("the Notch filter and possibly other filters"), the deep notch is cascaded with the Eq. 4 lead-lag, the partner swept over the same ranges: 67/75 combinations stabilise the loop. The best one ($\omega_x = 22.9$, $\zeta_{x,N} = 0.10$, $\zeta_{x,D} = 0.40$, selected for maximum delay margin) is stable but with clearly thinner nominal margins ($PM = 8.0°$). Its payoff is the bending-frequency robustness shown below.

Table 2: Task 2 — bending-filter trade at the fixed Task-1 gains (before the full-loop re-tune). Margins classified by frequency band.

Candidate	Aero $\lvert GM\rvert$ [dB]	Rigid PM [°]	Rigid $\lvert GM\rvert$ [dB]	$\lvert L(\omega_{BM})\rvert$ [dB]	DM [ms]	stable
B (no filter)	6.25	26.1	—	+29.0	—	no
C-LL (lead-lag)	5.96	11.4	8.02	+23.0	15	yes
C-N (deep notch)	6.08	14.6	10.43	−21.9	98	**yes**
C-T (triplet)	—	−7.3	—	−55.8	—	no
C-NLL (notch+LL)	5.72	8.0	7.41	−28.1	54	yes

3.3 Retained design and PD re-tuning

I keep the **deep notch**: it gain-stabilises the resonance ($|L(\omega_{BM})| = -22$ dB) and preserves the low-frequency aerodynamic gain margin. But with the Task-1 gains — placed on the *ideal*-actuator loop — the combined phase lag of the second-order TVC, the 20 ms transport delay and the notch has eaten most of the rigid phase margin: it collapses from 30° to **14.6**°, with the delay margin down at the $\sim$ 100 ms guideline (Table 2, C-N). The design must therefore be **re-tuned on the full loop**: raising the derivative gain restores the phase lost to the lag. Solving for the 6 dB / 30° target on the full open loop gives

$$K_{P,\theta} = 1.73, \qquad K_{D,\theta} = 0.69 \tag{11}$$

(the derivative gain up from the Task-1 0.44), which recovers a rigid phase margin of 30° at 3.2 rad/s and a delay margin of 165 ms, with the bending still gain-stabilised at $|L(\omega_{BM})| = -18$ dB — an 18 dB bending gain margin, above the ≥ 12 dB usually required of gain-stabilised modes [1]. Figure 10 shows the recovery on the Nichols chart, and Figure 11 the retained full-loop margins.

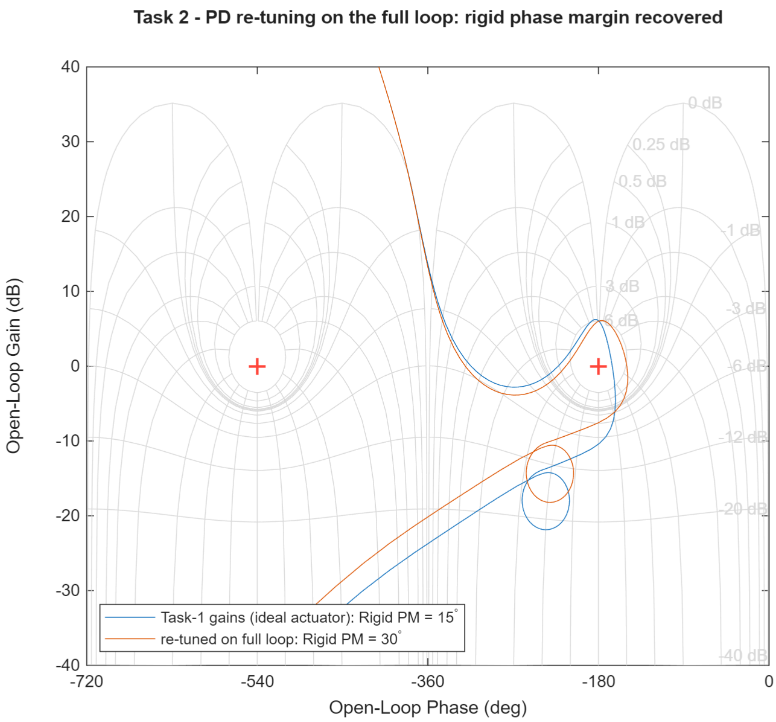


Figure 10: Task 2 — PD re-tuning on the full loop. With the Task-1 gains (ideal actuator) the actuator + delay + notch lag drives the loop close to the −180° critical point (rigid $PM = 15°$, blue); re-tuning on the full loop pulls the curve away and restores $PM = 30°$ (orange).

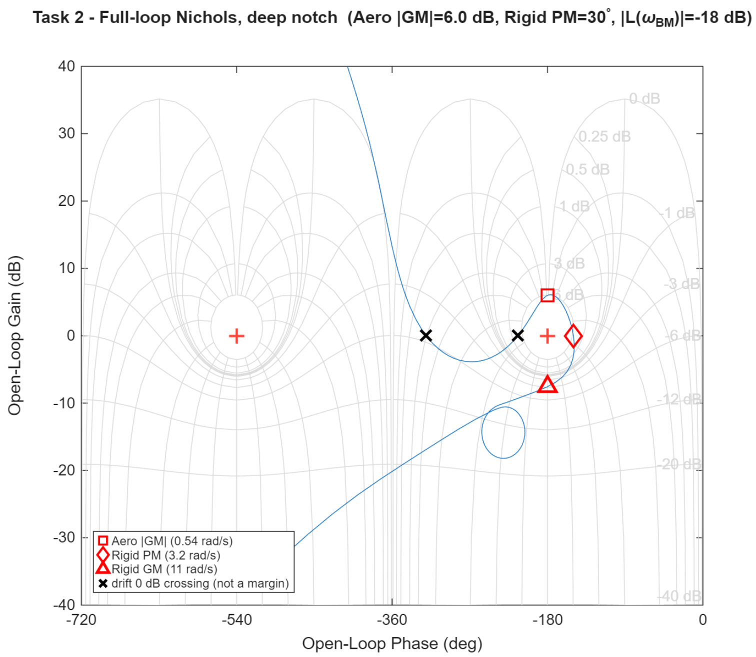


Figure 11: Task 2 — retained full-loop Nichols (deep notch, re-tuned PD); critical point at $(-180°, 0\text{ dB})$: aerodynamic gain margin ($|GM_a| = 6$ dB), rigid phase margin (30°) and the high-frequency rigid gain margin (set by the actuator lag) are marked. The bending mode is gain-stabilised, so its lobe stays well below 0 dB near ω_{BM} rather than threading the critical point.

The filter trade itself is shown on Figure 12: without compensation the +29 dB resonance destabilises the loop (red); the best Eq.-4 lead-lag alone (green) now barely stabilises the milder peak but with a thin margin; the deep notch drives the bending lobe far below 0 dB (blue).

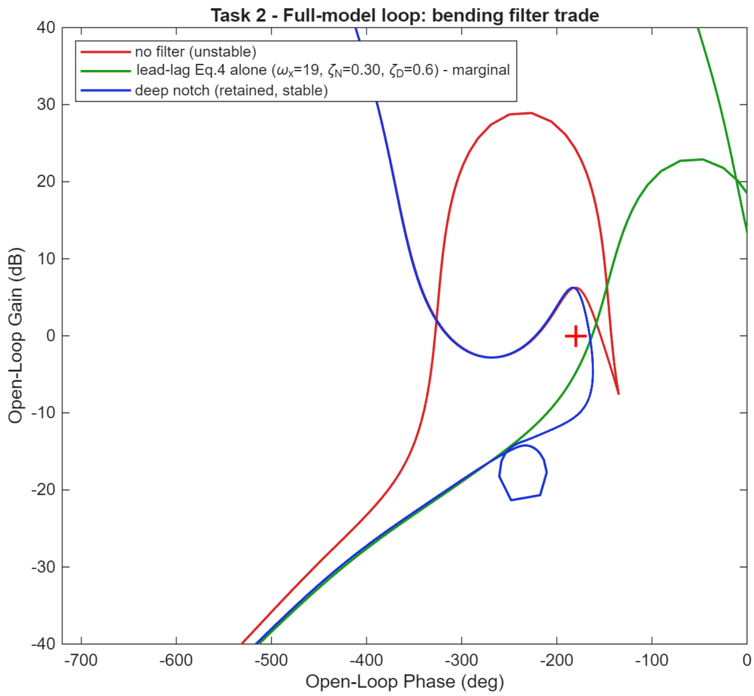


Figure 12: Task 2 — bending-filter trade on the Nichols chart (critical point at $-180°$; a common phase shift aligns the three loops at the rigid crossover): without a filter the +29 dB bending lobe (red) wraps over the critical point and the loop is unstable; the best Eq.-4 lead-lag alone (green) rotates the lobe but only marginally; the deep notch (blue) pushes the bending lobe far below 0 dB (the small loop near -18 dB), gain-stabilising it.

The s-plane view of Figure 13 makes the mechanism explicit: without the notch the closed-loop bending poles are pushed into the right half-plane by the +29 dB resonance and the loop is unstable, whereas the deep notch drops the loop gain at ω_{BM} and returns them to their lightly damped, stable open-loop location just left of the imaginary axis. The notch gain-stabilises the mode; it does not add damping to it.

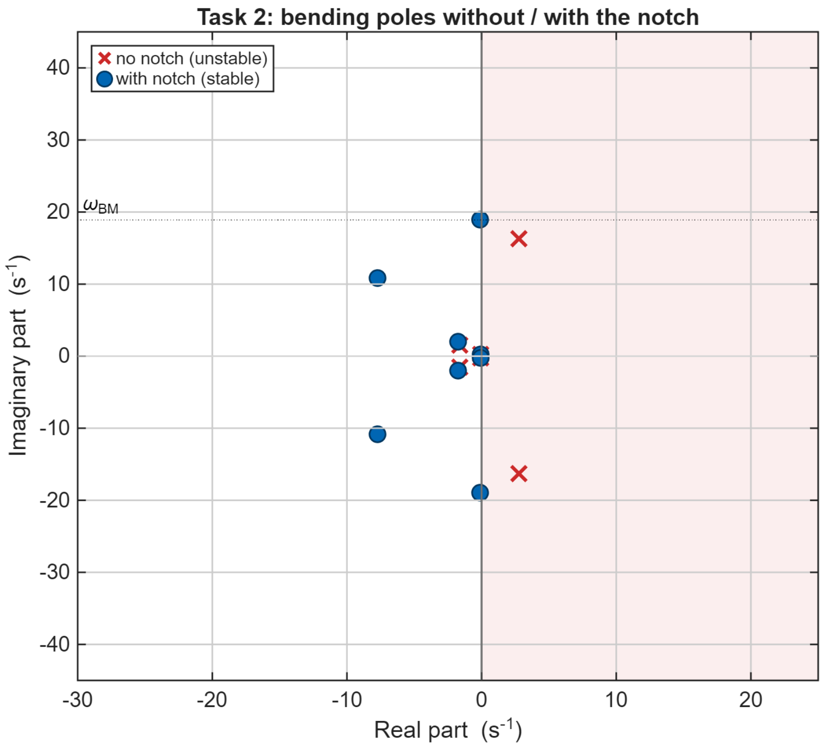
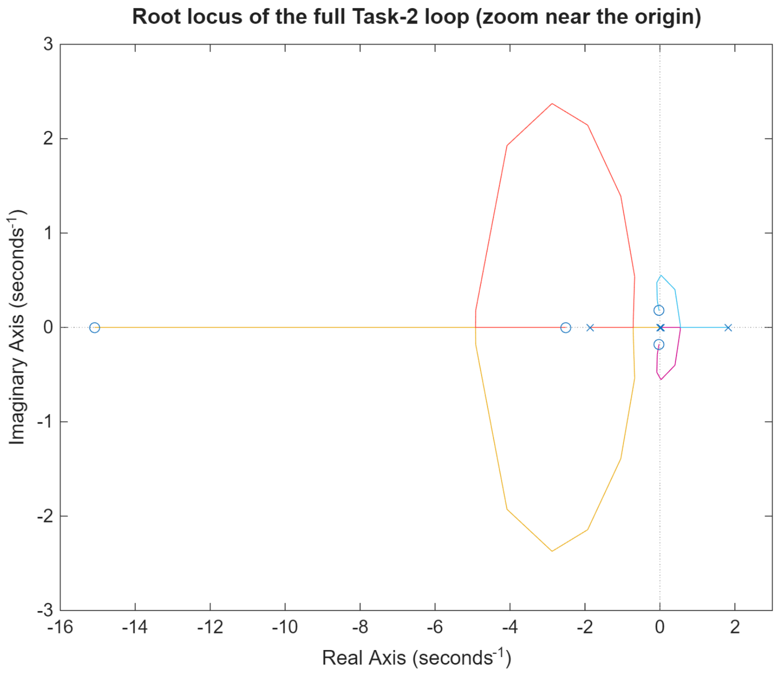
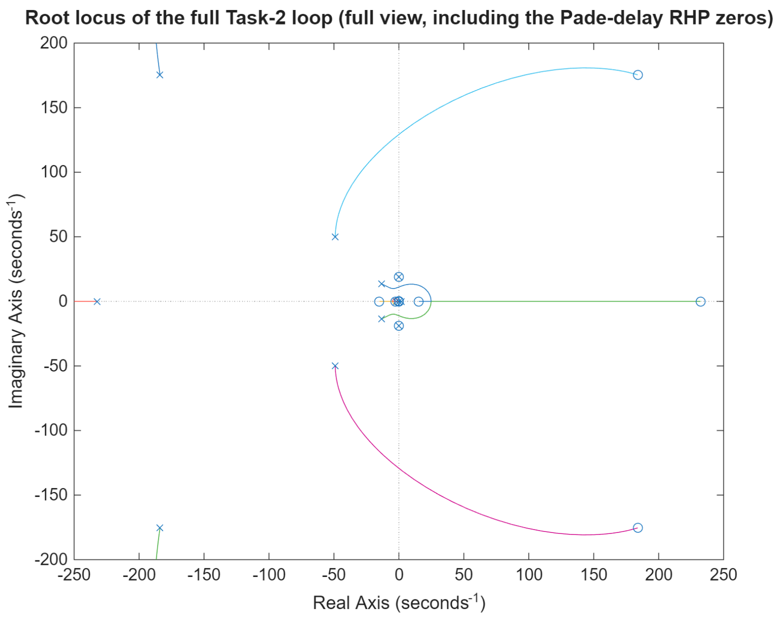


Figure 13: Task 2 — closed-loop bending poles without the notch (crosses, in the right half-plane) and with the notch (circles, returned to the stable neighbourhood of ω_{BM}).

The root locus of the full loop (Figure 14), from the built-in `rlocus`, tells the same story as a gain sweep. Figure 14(a) zooms on the low-frequency rigid and lateral-drift dynamics; near ω_{BM} the bending branch skirts the imaginary axis while the notch keeps the closed-loop poles ($k = 1$) just inside the left half-plane (Figure 13); and its far branches race toward the right-half-plane zeros of the Padé delay approximation, resolved in the full-scale view of Figure 14(b).



(a) Zoom near the origin.



(b) Full view, including the Padé-delay RHP zeros.

Figure 14: Task 2 — root locus of the full loop (flexible plant, TVC servo, Padé delay and notch; $\times$: open-loop poles, $\circ$: open-loop zeros). The design is stable, with all closed-loop poles in the left half-plane. (a) Low-frequency detail near the origin: the rigid-body and lateral-drift dynamics, the unstable aerodynamic pole at $+1.8$ s^{-1} and the real zeros at -2.5 and -15 s^{-1}; the bending poles ($\omega_{BM} \approx 18.9$ rad/s), the notch and the far delay zeros fall outside this window. (b) The same locus zoomed out: the far branches terminate on the transport-delay RHP zeros at $+232$ and $+184 \pm 175i$ s^{-1} (mirror LHP poles at -232 and $-184 \pm 175i$).

3.4 Sensitivity to the bending-frequency knowledge

The deep notch pays for its performance with a -51 dB null only 0.08 rad/s wide, i.e. it assumes ω_{BM} is known exactly — true in this exercise, where the LPV dataset provides it, but never true in practice. Table 3 makes the trade explicit: each filter is held fixed (with the re-tuned PD) while the true plant bending frequency is perturbed. The deep notch tolerates -10% but goes unstable at $+5\%$ — the asymmetric edge of its narrow null; the notch+lead-lag combination, by phase-stabilising whatever leaks past the detuned notch, holds the full $\pm 10\%$ band at the cost of the thin nominal margins seen above. In a flight design, where ω_{BM} drifts with propellant depletion, the robust variant (or a phase-stabilised loop) would be mandatory; for this max-$\bar{q}$ snapshot I keep the exact notch.

Table 3: Task 2 — closed-loop stability with the filters fixed (re-tuned PD) and the true bending frequency perturbed by up to $\pm 10\%$.

Candidate	$0.90\,\omega_{BM}$	$0.95\,\omega_{BM}$	ω_{BM}	$1.05\,\omega_{BM}$	$1.10\,\omega_{BM}$
C-N — deep notch	yes	yes	yes	no	no
C-NLL — notch+lead-lag	yes	yes	yes	yes	yes

3.5 Validation and comparison with the rigid model

The gust response of the stabilised full model is shown in Figure 15; all four assignment time histories remain well damped, with peaks $\theta = 0.23°$, $z = 2.27$ m, $\delta = 0.51°$ and a load indicator $\bar{q}\,\alpha = 45.8$ kPa deg (peak incidence $\alpha = 0.565°$), essentially the rigid case. Because the notch is localised at the bending frequency and the TVC bandwidth (70 rad/s) is well above the control crossover, the full-model response is almost indistinguishable from the rigid one (Figure 16): the added dynamics affect stability robustness (the bending and delay margins) far more than nominal performance.

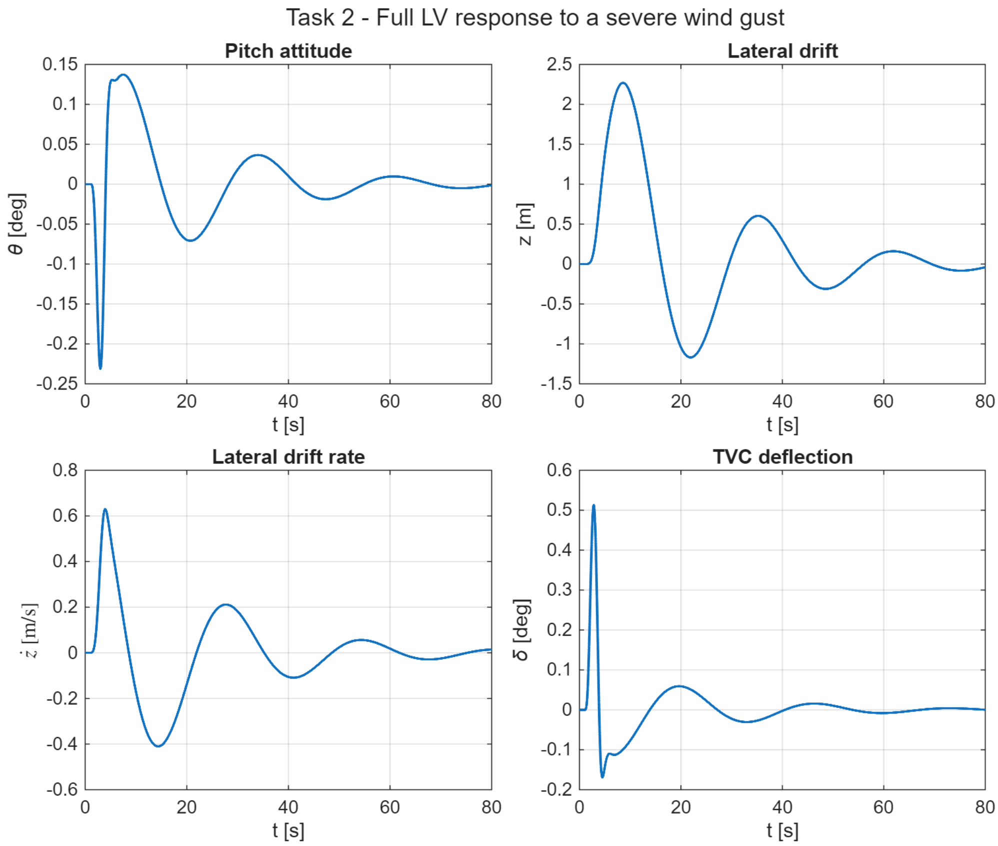


Figure 15: Task 2 — full-model response to the severe wind gust: pitch attitude θ, lateral drift z, drift rate $\dot{z}$ and TVC deflection δ.

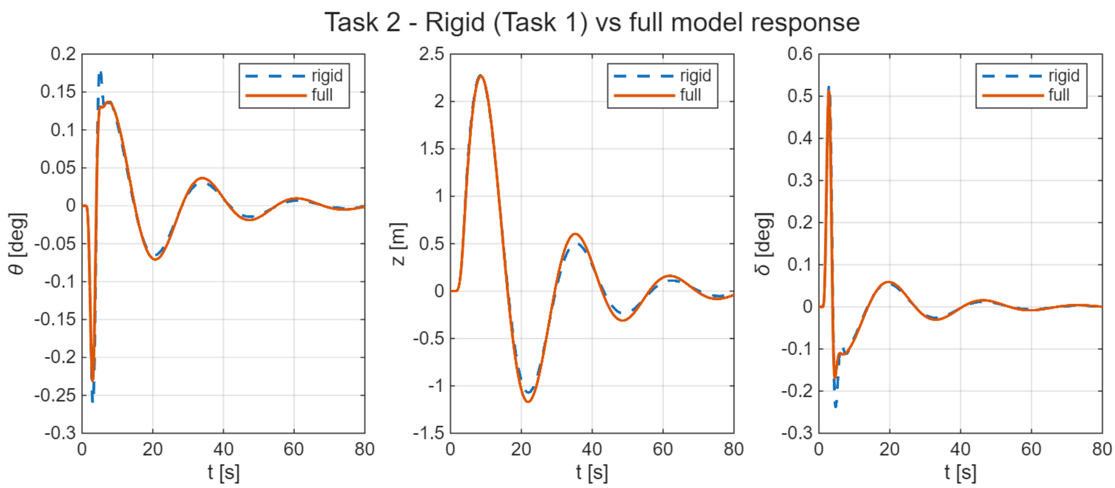


Figure 16: Task 2 — rigid (Task 1) versus full-model gust response: pitch attitude, lateral drift and TVC deflection are nearly coincident.

3.6 A note on the tail-wags-dog effect

A second flexible effect bounds what any filter can achieve. The gimballed nozzle has its own inertia, so its angular acceleration reacts back on the airframe. Following the course treatment [2, Lecture 20], the oscillating nozzle exerts on the vehicle a lateral reaction force and a moment

$$R = -\, m_E \ell_E \, \ddot{\delta}, \qquad M_{TWD} = \left(I_E + m_E \ell_E^2\right) \ddot{\delta}, \tag{12}$$

proportional to the gimbal acceleration $\ddot{\delta}$ (m_E nozzle mass, ℓ_E pivot-to-nozzle-CG distance, I_E nozzle inertia). This **tail-wags-dog** (TWD) coupling feeds the second derivative of the control back into the plant; it adds phase lag and an effective transmission zero that caps the achievable closed-loop bandwidth. The $-\phi_{1,TVC} T_c$ entry in the bending-forcing column of Eq. (1) is the flexible-mode analogue of this nozzle-reaction term. My model treats the actuator as a stiff second-order block (Eq. (4)) and so neglects the nozzle back-reaction — the dog-wags-tail feedback, which would add the gimbal angle as an extra state. A higher-fidelity design would keep it, one of the extensions noted in the conclusions.

4 Task 3 — Robustness to Parametric Uncertainty

The aerodynamic moment $\mu_\alpha = A_6$ and the control effectiveness $\mu_c = K_1$ are treated as uncertain. The controller designed in Task 2 (PD gains plus bending notch) is held **fixed** and I assess its robustness over the four corner cases of the uncertainty box, i.e. the vertices V1–V4 obtained by independently varying μ_α and μ_c by $\pm 30\%$. Since each vertex perturbs both parameters at once, I also evaluate the four one-at-a-time variations S1–S4 ($\pm 30\%$ on a single parameter): they sit on the edge midpoints of the box rather than on its corners, but they attribute the observed effects to each parameter.

4.1 Frequency- and time-domain analysis

Table 4 collects, for every case, the classified rigid-body margins — the aerodynamic (low-frequency) gain margin, the rigid phase margin, and the high-frequency rigid gain margin — the delay margin, and the peak pitch and lateral-drift excursions to the same severe gust. All vertices stay stable: the fixed re-tuned Task-2 controller tolerates the entire $\pm 30\%$ uncertainty box without re-design.

Table 4: Task 3 — classified margins and gust-response peaks: box vertices (V1–V4, the assignment corner cases) and one-at-a-time sensitivities (S1–S4).

Case	μ_α	μ_c	Aero $\lvert GM\rvert$ [dB]	Rigid PM [°]	Rigid $\lvert GM\rvert$ [dB]	DM [ms]	peak θ [°]	peak z [m]
Nominal	1.00	1.00	6.00	30.0	7.56	165	0.231	2.27
V1	0.70	0.70	5.70	26.6	8.70	196	0.241	2.28
V2	0.70	1.30	11.06	30.8	6.50	121	0.097	1.57
V3	1.30	0.70	**0.91**	**18.0**	8.75	209	**0.878**	**4.92**
V4	1.30	1.30	6.17	30.9	6.57	135	0.222	2.26
S1	0.70	1.00	8.79	30.6	7.53	156	0.142	1.80
S2	1.30	1.00	3.93	28.9	7.59	176	0.346	2.83
S3	1.00	0.70	2.95	23.6	8.72	206	0.417	3.19
S4	1.00	1.30	8.26	30.9	6.54	128	0.154	1.90

The Nichols overlay over the nominal case and the four vertices (Figure 17) shows the corner loops circling the $-180°$ critical point: V2 rides highest (largest aerodynamic gain margin) while **V3** rides closest to the critical point (its aerodynamic margin nearly vanishes). The time-domain overlay (Figure 18) shows the corresponding spread of the gust response.

4.2 Discussion

The sensitivities S1–S4 attribute the effects. The **aerodynamic gain margin** is set by the control-to-aero authority ratio $K_1 K_{P,\theta}/A_6$: it erodes when μ_α grows (S2: $6 \to 3.9$ dB) or μ_c shrinks (S3: $6 \to 3.0$ dB) — both reduce that ratio and scale up the gust excursions — and improves in the opposite corners. Raising μ_c helps the aerodynamic margin (S4: 8.3 dB) but adds loop gain at high frequency, which lowers the *rigid* gain margin and the delay margin (S4: 6.5 dB, 128 ms): the two-sided gain band once more. The vertices combine these mechanisms:

- **V3** ($\mu_\alpha \uparrow$, $\mu_c \downarrow$) is the worst case, because the two effects compound: the airframe is more unstable and the authority to fight it is reduced at once. The aerodynamic gain margin nearly vanishes (0.9 dB) with $PM = 18°$, and the gust excursions grow to 0.88° in pitch (more than double the worst one-at-a-time case, S3) and 4.9 m of drift (about 1.5× S3). This is why the actual vertices, not only the single-parameter variations, must be tested.
- **V2 and V4** ($\mu_c \uparrow$) have the smallest high-frequency margins (rigid $|GM| \approx$ 6.5 dB, $DM \approx$ 120–135 ms), but these stay comfortable: with the re-tuned, deeply gain-stabilised bending channel the flexible side is no longer the binding constraint.
- **V1** ($\mu_\alpha\downarrow$, $\mu_c\downarrow$) is benign: the two reductions nearly cancel, leaving margins close to nominal.

So the binding robustness limit is the *aerodynamic* gain margin, driven down by $\mu_\alpha \uparrow$ / $\mu_c \downarrow$; the worst-case combination V3 leaves it at $\sim$ 1 dB. Because the margins are non-monotonic in μ_c (it helps the aerodynamic margin while lowering the high-frequency one), the global worst case over the ±30% box need not sit on a vertex: the corners and the one-at-a-time sensitivities are only a screening. The fixed classical design survives the whole box, but a requirement of, say, ≥ 3 dB of aerodynamic gain margin in every corner would already be violated by V3 and would call for either gain scheduling on $\bar{q}$ or a robust-synthesis retune — the standard routes beyond a single fixed classical design.

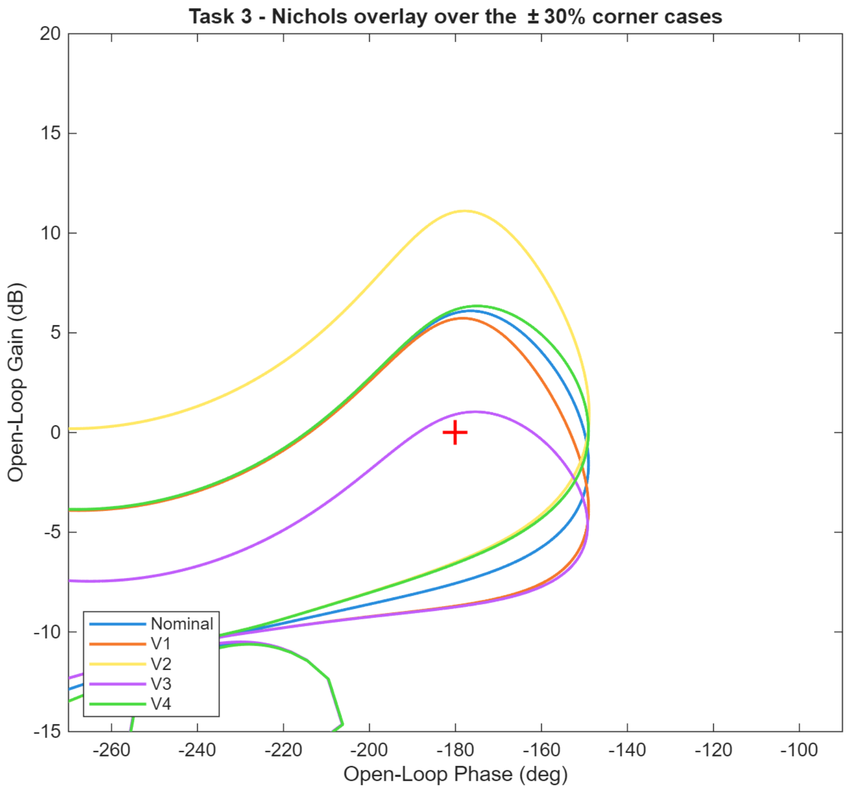


Figure 17: Task 3 — Nichols overlay over the nominal case and the four uncertainty-box vertices (critical point at $-180°$). Every corner stays clear of the critical point; V3 (max instability, min authority) rides closest, its aerodynamic gain margin nearly vanishing.

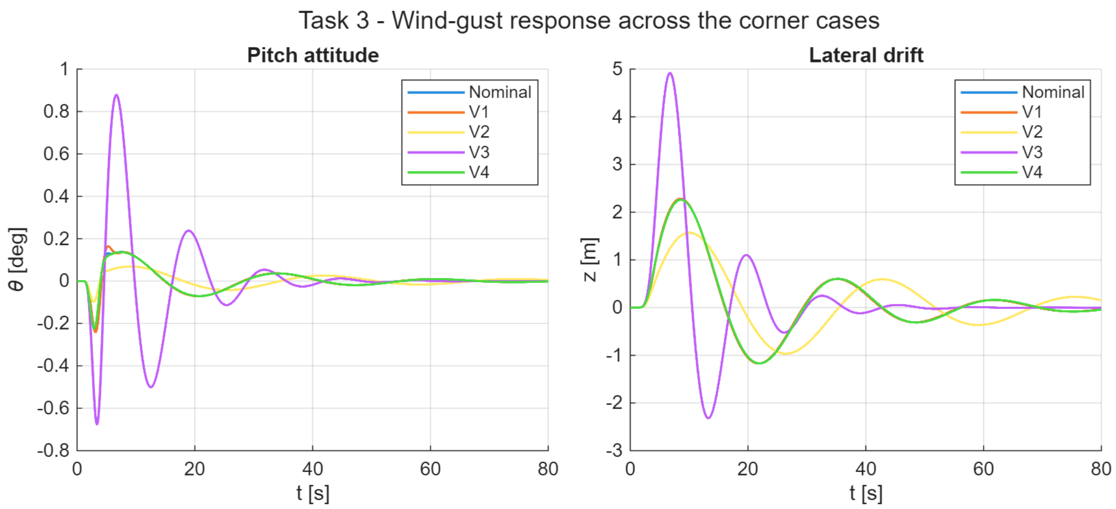


Figure 18: Task 3 — wind-gust response over the nominal case and the four vertices: pitch attitude (left) and lateral drift (right).

Figure 19 closes the frequency-domain view in the s-plane: the closed-loop poles for the nominal case and the four vertices all stay in the left half-plane, so the fixed classical design is robustly stable over the whole $\pm 30\%$ box. The slow lateral-drift pair near the origin sits closest to the imaginary axis (the near-neutral drift mode of the max-$\bar{q}$

design), while the notched bending poles near ω_{BM} stay comfortably damped.

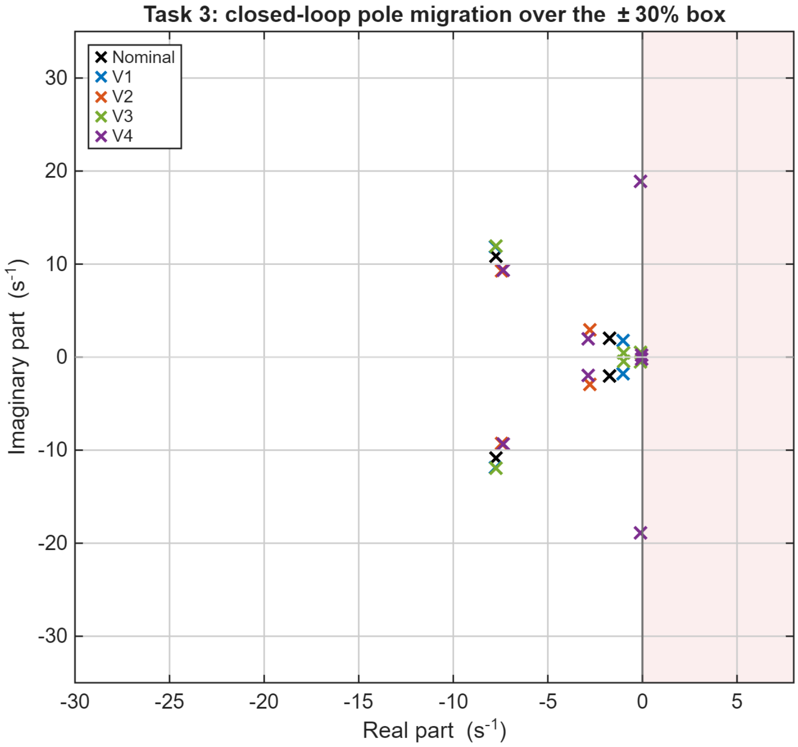


Figure 19: Task 3 — closed-loop pole migration over the nominal case and the four ±30% vertices (zoom near the imaginary axis); all poles remain in the left half-plane.

5 Conclusions

I designed and validated a classical attitude control system for an aerodynamically unstable launch vehicle at the max-$\bar{q}$ condition, working entirely in the frequency domain.

- **Task 1.** A proportional–derivative attitude law with weak negative drift feedback stabilises the rigid airframe (open-loop pole at +1.84 rad/s) and meets the design targets $|GM_a| \approx 6$ dB, $PM \approx 30°$ — the aerodynamic gain margin and rigid-body phase margin, read on the full conditionally-stable loop and classified by frequency band.
- **Task 2.** Introducing the TVC actuator, the transport delay and the lightly damped bending mode destabilises the loop through a +29 dB resonance. A four-way filter trade retained a deep gain-stabilising notch at ω_{BM} (the Eq.-4 lead-lag alone now stabilises the milder peak in 45/75 cases, but only marginally, while a notch triplet over-notches and goes unstable at nominal). Crucially, the actuator, delay and notch lag collapse the rigid phase margin of the ideal-actuator Task-1 gains from 30° to 15°, so the PD is **re-tuned on the full loop** ($K_{D,\theta} : 0.44 \to 0.69$), recovering 30° and a 165 ms delay margin with the bending gain-stabilised at −18 dB. The deep notch leans on knowing the bending frequency: it tolerates −10% but fails at +5%, whereas a notch+lead-lag variant holds the full ±10% at the price of thinner nominal margins.
- **Task 3.** The fixed re-tuned Task 2 controller is robust to ±30% independent variations of μ_α and μ_c: all nine cases (the four box vertices plus the four single-parameter edges) remain stable. The binding limit is the aerodynamic gain margin (the control-to-aero authority ratio): it erodes as μ_α grows or μ_c shrinks, and the worst vertex V3 (high μ_α with low μ_c) compounds both, leaving it at only 0.9 dB with 18° of phase and doubling the gust excursions — the quantitative argument for gain scheduling in a flight design.

Limitations and extensions. Four modelling choices bound the scope of this design. (i) The pure proportional–derivative law leaves a steady-state attitude error to a constant gimbal misalignment, which a proportional–integral–derivative extension would null. (ii) The stiff second-order TVC model neglects the nozzle back-reaction (the dog-wags-tail feedback) and the hydraulic load-resonance of the actuator; retaining the gimbal angle as a state, together with the tail-wags-dog reaction of Eq. (12) [2, Lecture 20], is the natural next fidelity step. (iii) Sensor placement relative to the bending-mode nodes matters as much as the filter — an aft gyro station flips the sign of the modal slope and can phase-stabilise a mode that would otherwise need a notch [1, Sec. 3.3.2.2] — whereas here the INS station is fixed and only the filter is traded. (iv) Propellant sloshing is omitted; it would add low-frequency, lightly damped modes coupling into the rigid loop and the drift channel.

The design shows the textbook trade-off of launch-vehicle attitude control [2, Lecture 18]: enough gain to stabilise the unstable airframe, but not so much that it excites the structural modes, and the Nichols chart lets me see both constraints on one plot. A Simulink block-diagram mirror of the closed loop reproduces the script results and is documented in the accompanying `models/SIMULINK_GUIDE.md`.

A Implementation: Key Code Excerpts

The full MATLAB sources live in the `HM3/` folder of the repository: `main_task1.m`, `main_task2.m` and `main_task3.m` drive the three tasks, and the helper functions (`load_hw3_params`, `build_plant_rigid`, `build_plant_full`, `build_tvc`, `build_notch_filter`, `design_controller`, `assemble_loop`, `simulate_gust_response`) are shared across them. This appendix collects the three routines that set up the control problem: the rigid plant, the loop closure the Nichols analysis is drawn from, and the bending notch.

Listing 1 builds the rigid pitch-plane state-space model at the max-$\bar{q}$ condition by dropping the bending states from the assignment dynamics; the unstable airframe pole sits at $+\sqrt{A_6}$.

```
function G = build_plant_rigid(p)
% Rigid pitch-plane LV plant at max-qbar (Task 1, Eq. 1 minus bending).
%   INPUT
%     p - param struct (load_hw3_params)
%   OUTPUT
%     G - ss, 4 states [z zdot theta thetadot], in [delta alpha_w],
%         out [theta_m thetadot_m z_m zdot_m theta z zdot]
%   First 4 outputs feed the controller (= true states here, no bending);
%   last 3 are plotting signals.

A = [0   1        0          0;
     0   p.a1     p.a1*p.V+p.a4   0;
     0   0        0          1;
     0   p.A6/p.V p.A6       0];

Bd = [0; p.a3; 0; p.K1];          % delta column
Bw = [0; -p.a1*p.V; 0; -p.A6];    % alpha_w column

% Outputs: [theta_m, thetadot_m, z_m, zdot_m, theta, z, zdot]
%            state index:   z=1  zdot=2  theta=3  thetadot=4
C = [0 0 1 0;    % theta_m    = theta
     0 0 0 1;    % thetadot_m= thetadot
     1 0 0 0;    % z_m        = z
     0 1 0 0;    % zdot_m     = zdot
     0 0 1 0;    % theta
     1 0 0 0;    % z
     0 1 0 0];   % zdot
D = zeros(7, 2);

G = ss(A, [Bd Bw], C, D);
G.StateName  = {'z','zdot','theta','thetadot'};
G.InputName  = {'delta','alpha_w'};
G.OutputName = {'theta_m','thetadot_m','z_m','zdot_m','theta','z','zdot'};
G.Name = 'Plant_Rigid';
end
```

Listing 1: Rigid pitch-plane LV plant (`build_plant_rigid.m`).

Listing 2 closes the PD attitude and drift loop around the plant, optionally inserting the TVC, delay and notch chain through `Wact`, and returns the SISO open-loop transfer broken at the TVC deflection. This open-loop transfer is what the Nichols charts in

Tasks 1–3 are built from.

```

% --- open loop at the break point 'delta' (1+L convention) ---
L = getLoopTransfer(T, 'delta', -1);
L = minreal(tf(L), 1e-6);

info = struct('Kc', Kc, 'Wact', Wa);
end
```

Listing 2: Loop closure and open-loop transfer for the Nichols analysis (`assemble_loop.m`).

Listing 3 is the second-order lead-lag / notch section used in Task 2 to stabilize the lightly damped first bending mode.

```
function Hx = build_notch_filter(wx, zN, zD, numSign)
% Lead-lag / notch section for bending stabilisation (Eq. 4).
%   Hx(s) = (s^2 + sgn*2*zN*wx*s + wx^2) / (s^2 + 2*zD*wx*s + wx^2)
%   INPUT
%     wx      - centre frequency [rad/s] (~ wBM +/- 4)
%     zN      - numerator damping (guideline 0.1-0.3)
%     zD      - denominator damping (guideline 0.4-0.6)
%     numSign - -1 (default, Eq. 4 as printed): RHP zeros, non-min-phase
%               phase-shaper; +1: min-phase symmetric notch
%   OUTPUT
%     Hx - tf

arguments
    wx (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
    zN (1,1) {mustBeNumeric, mustBeReal, mustBeNonnegative}
    zD (1,1) {mustBeNumeric, mustBeReal, mustBePositive}
    numSign (1,1) {mustBeMember(numSign, [-1, 1])} = -1
end

num = [1, numSign*2*zN*wx, wx^2];
den = [1,         2*zD*wx, wx^2];

Hx = tf(num, den);
Hx.Name = 'Notch_Hx';
end
```

Listing 3: Lead-lag / notch filter for bending stabilization (`build_notch_filter.m`).

References

[1] A. L. Greensite. Analysis and design of space vehicle flight control systems. volume vii — attitude control during launch. Technical Report CR-826, NASA, Washington, DC, July 1967.

[2] A. Zavoli. Dynamics and control of launch vehicles — course lecture notes. Graduate course, Sapienza University of Rome, 2026. Academic Year 2025/26.

[3] Timothy M. Barrows and Jeb S. Orr. *Dynamics and Simulation of Flexible Rockets*. Academic Press, 2020.